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FA23-BAI-066

Topics in AI (Assignment # 01)
Custom Reinforcement Learning Agent

1. Problem Description

Taxi-v4 is a discrete 5 x 5 navigation problem. The taxi must travel to one of four passenger locations, pick up the passenger, and deliver them to the indicated destination. The observation is an encoded integer with 500 possible states: taxi row (5), taxi column (5), passenger location (5, including inside taxi), and destination (4). There are 6 actions: south, north, east, west, pickup, and dropoff.

The environment gives -1 per ordinary time step, -10 for an illegal pickup/dropoff, and +20 for correct delivery. Success means the episode terminates after delivery; a 200-step limit truncates incomplete episodes.

2. Methodology

A tabular Q-learning agent was implemented from scratch. At each step, the epsilon-greedy policy either explores a uniformly sampled action or exploits a maximum-Q action. The update is:

Q(s,a) <- Q(s,a) + alpha [r + gamma max Q(s',a') - Q(s,a)].

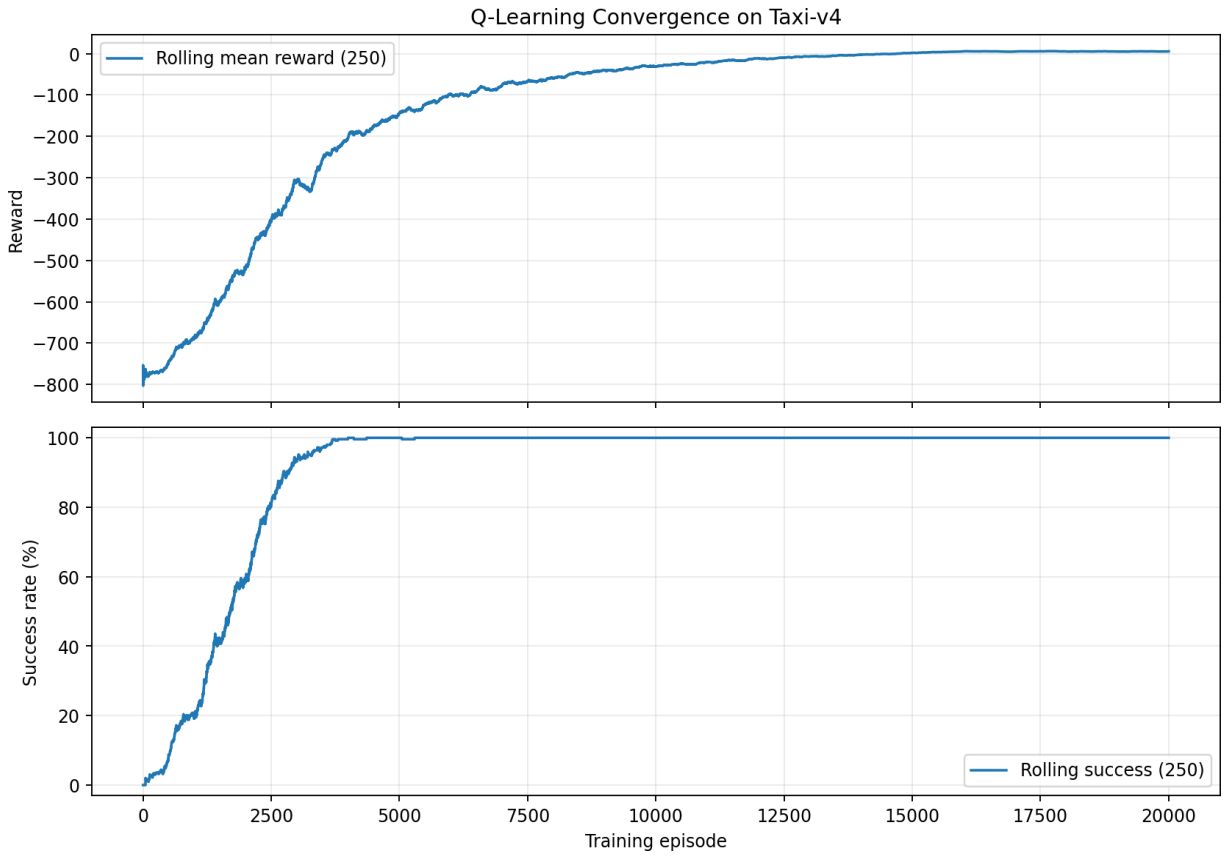
For a true terminal transition, the bootstrap term is zero. At a time-limit truncation, the learner still bootstraps, but stops the episode. Exploration decreases linearly toward its floor.

Implementation: **agent.py** contains training, exploration, and evaluation; **utils.py** produces plots, the real RGB-rendered GIF, and this report; **main.py** runs and exports the experiment. Only NumPy is used to represent and update the learned Q-values.

3. Experimental Setup

Setting	Value	Setting	Value
Environment	Taxi-v4	Episodes	20000
Learning rate (alpha)	0.7	Discount (gamma)	0.95
Start epsilon	1.0	End epsilon	0.05
Decay fraction	0.8	Random seed	42
Eval episodes	100	Q-table shape	500 x 6

4. Results: Convergence



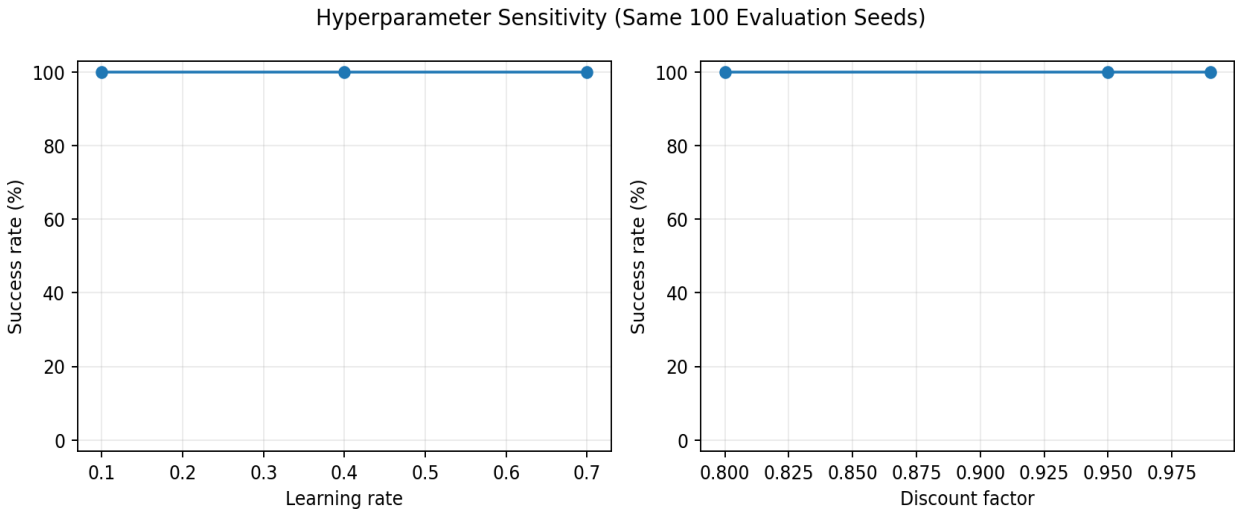
Average training reward over the first 250 episodes was **-770.99**; over the last 250 it was **5.36**. The rolling curves indicate whether learning stabilized and how reliably the taxi completes trips.

5. Evaluation Against Random Policy

Policy	Mean reward	Success	Mean steps
Untrained random	-768.47	1.0%	199.0
Trained greedy	8.05	100.0%	12.9

Both policies use the same 100 evaluation starting seeds. No epsilon-greedy exploration is used for the trained policy during evaluation. Success is the proportion ending in legal passenger delivery.

6. Hyperparameter Sensitivity



Parameter	Value	Mean reward	Success
alpha	0.1	8.05	100.0%
alpha	0.4	8.05	100.0%
alpha	0.7	8.05	100.0%
gamma	0.8	8.05	100.0%
gamma	0.95	8.05	100.0%
gamma	0.99	8.05	100.0%

In the shorter sensitivity runs, the highest ranked setting was **alpha=0.1** with mean evaluation reward **8.05** and success rate **100.0%**. These runs are shorter than the main training run and compare only the listed candidate values.

7. Discussion and Failure Modes

Initial episodes can accumulate many negative step rewards and illegal pickup/dropoff penalties. Training improves through repeated state-action updates. Remaining failures can arise from states that have been visited too rarely during training, too little exploration, or an unfavorable learning rate. A high success rate alongside shorter routes is evidence of useful learned behavior; the evaluation scores and plots above are the measured results rather than assumed outcomes.

8. Conclusion

This implementation learned a tabular policy for Taxi-v4. Across 100 evaluation episodes, the trained agent achieved **100.0%** successful deliveries and mean reward **8.05**, compared with **-768.47** for the random baseline. A successful demonstrated rollout took **14** steps. The exported Q-table and GIF enable direct inspection of the learned behavior.